

Bregman divergence and Hardy-Stein inequalities

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I will introduce the notion of Bregman divergence, which will be then used to obtain Hardy-Stein inequalities for semigroups related (mostly) to nonlocal Lévy-type operators. Time permitting, other applications - such as optimal Hardy inequalities, Douglas identities, extension and trace theorems etc. will be presented.

The talk will be partially based on the paper

Krzysztof Bogdan, Michał Gutowski, Katarzyna Pietruska-Pałuba, *Polarized Hardy-Stein identity*, J. Funct. Anal. **288** (2025), Paper No. 110827, 39 pp.